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Token Economics

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Abstract

Tokenomics, the economic framework governing digital tokens and the behavior of their users in blockchain networks, plays a crucial role in incentivizing participation and sustaining decentralized ecosystems. This entry explores key aspects of tokenomics, including token supply, distribution, and utility. We discuss governance mechanisms through token-based voting, various tokenomic strategies such as token burning, minting, and staking, and risk management and collateralization mechanisms. Additionally, we highlight how these mechanisms ensure network stability, promote decentralization, and align participant incentives. By analyzing real-world examples, this paper demonstrates the importance of tokenomics in shaping the sustainability and success of blockchain-based networks and decentralized finance (DeFi) protocols.

Keywords: Tokenomics, Digital Token, Cryptocurrency, Blockchain, Decentralization, Incentive System

Introduction

“Token economics,” also referred to as tokenomics, focuses on the economic aspects of digital tokens (e.g., cryptocurrencies such as BTC, governance tokens such as MKR in MakerDAO) used within blockchain networks (e.g., Bitcoin or Ethereum). Tokenomics specifically involves key design choices focusing on the creation, distribution, and management of digital tokens from a market economics standpoint. These design choices play an important role for incentivizing participation in blockchain networks, helping to sustain them through the strategic use of their native digital tokens that result in ongoing development and improvements (Ellinger et al., 2024). In this regard, tokenomics has also been referred to as an incentive system.

The fundamental idea behind tokenomics is thus to incentivize blockchain network participants to encourage desired behaviors, which is critical due to the absence of a central, managing authority in blockchain networks. In the absence of a central authority and classical command-and-control model, blockchain networks striving toward a high degree of decentralization require a strong incentive system to make participation and cooperation in the blockchain network viable and sustainable (Lesche et al., 2022). This could include staking rewards for validators who secure the network, governance rights for token holders, or other benefits that align participants' actions with the network's long-term health. As all of these incentive mechanisms involve the exchange and use of digital tokens, a key aspect of tokenomics is token supply and distribution.

Tokenomics involves decisions about the total supply of digital tokens, their issuance method, and distribution strategy. These decisions can influence whether a digital token has either inflationary aspects or deflationary aspects (e.g., Ripple's XRP token has both inflationary aspects due to ongoing escrow releases and deflationary aspects due to transaction fee burning and periodic escrow returns). The basis for these decisions is the purpose for which digital tokens were created. The purpose of a digital token is socially constructed by the participating blockchain community (sometimes, this purpose may be ill-defined and emergent, resulting in different perceptions of the community and the original designer's intention). Examples for purpose include the token acting as a medium of exchange or participating in collective decision-making via token-based voting. Some tokens grant access to certain services on the blockchain (utility tokens), while others may represent assets or equity in a project (security tokens) (Sockin & Xiong, 2023). In addition, many blockchain projects use digital tokens (governance tokens) to facilitate decentralized governance, allowing token holders to vote on key decisions affecting the project's future. A governance token is a digital object (token) that encodes rights of a stakeholder in a blockchain-based collective to participate in governance-related decision-making through voting and is used in combination with smart contracts on the blockchain to exercise these rights and make the participation process transparent and verifiable.

Regardless of their specific use case and purpose, digital tokens are bought and sold on crypto exchanges, with the price mechanism directing the activities of market participants (sellers, buyers, market makers). Digital tokens are thus subject to market dynamics, affecting tokenomics design choices. The liquidity, trading volume, and market capitalization of digital tokens are vital for their viability. A role played by tokenomics is to ensure that digital tokens are sufficiently liquid to facilitate trading and price discovery without excessive volatility.

In sum, tokenomics is not just about creating digital tokens but also about crafting a sustainable economic model that supports a decentralized blockchain-based application or blockchain network as a whole. By carefully designing tokenomics, blockchain projects and network leaders can ensure that all participants in the ecosystem are incentivized to act in ways that increase the network's collective value and sustainability. This includes ensuring that tokens have real utility, aligning incentives with user and participant behavior, and managing supply to prevent issues like inflation or deflation from undermining the token's value. Moreover, tokenomics plays

a critical role in the broader “token economy,” which refers to the economic systems and interactions that are enabled by digital tokens. These tokens facilitate the exchange of value and services within a blockchain network, potentially creating a self-sustaining economic ecosystem that can operate independently of traditional financial systems (this depends heavily on the effectiveness of the tokenomics decisions and design choices, as well as external factors such as market dynamics).

There are a number of tokenomics mechanisms used across different blockchain projects. In what follows, we highlight some of the most salient ones.

Initial Distribution

The way digital tokens on a blockchain are initially distributed, such as through Initial Coin Offerings (ICOs), private sales, or airdrops, can significantly impact their market dynamics and perceived value, thereby shaping the tokenomics system. ICOs enable startups to bypass traditional financial intermediaries like banks and venture capitalists, allowing them to raise funds directly from investors. This democratizes the investment process, as almost anyone can participate in a public ICO. The tokens sold during an ICO can serve various purposes. They might provide access to a product or service, act as a medium of exchange within a specific ecosystem, or even offer governance rights within a platform. The specific utility of the token is a central component of its tokenomics.

For example, Ethereum’s initial token distribution through its ICO in 2014 significantly shaped its tokenomics and future success. The public sale of Ether (ETH) tokens allowed for widespread participation, creating a large and diverse community of stakeholders from the outset. This broad distribution of tokens laid the foundation for a more decentralized governance model, where a wide range of participants have a vested interest in the network’s success. The Ethereum Foundation conducted a public sale where investors could purchase ETH tokens using Bitcoin. This approach granted widespread access to ETH, allowing a wide range of participants to acquire tokens. The ICO raised approximately 31,000 BTC (worth about \$18.3 million at the time) in exchange for about 60 million ETH (see, e.g., <https://blockworks.co/price/eth>). The team allocated 12 million ETH to the development fund, while the remaining tokens were distributed to ICO participants. This balance between community participation and development support by the Ethereum Foundation has been a key factor in shaping Ethereum’s current governance model, which relies on a blend of decentralized input and core developer leadership to evolve the platform. The initial token distribution has also influenced Ethereum’s approach to decision-making. Unlike some other blockchain projects, Ethereum has chosen to maintain an off-chain governance model, avoiding on-chain voting systems that could potentially be dominated by large token holders. Instead, Ethereum relies on a more nuanced process of Ethereum Improvement Proposals (EIPs), which allows for extensive discussion and consideration before implementation.

In sum, this initial distribution laid the groundwork for Ethereum's decentralized ecosystem, providing a broad base of token holders who could participate in network activities, development, and governance.

Token Burning

Token burning is an important mechanism related to tokenomics. It involves the permanent removal of a certain number of tokens from circulation. This is typically achieved by sending the tokens to a specific, unspendable address, often referred to as a "burn address," where they cannot be accessed or used again. By reducing the total supply of tokens, burning can create scarcity, which may lead to an increase in the value of the remaining tokens if demand remains constant or increases. Token burning can potentially also drive up the price of a cryptocurrency or digital token by creating a deflationary effect. Regular token burns can signal to investors that a project is committed to increasing token value over time. This can encourage investors to hold onto their tokens longer, contributing to market stability and reducing volatility.

A notable example of token burning with a significant impact is the Binance Coin (BNB) burn conducted by Binance. Binance has implemented a systematic quarterly token burn process, where they burn a portion of BNB tokens based on their quarterly profits. This process is aimed at reducing the total supply of BNB, thereby potentially increasing its value over time.

The BNB token burn is one of the most well-known and impactful examples in the cryptocurrency space. Binance commits to burning BNB tokens until 100 million BNB are removed from circulation, which represents 50% of the total supply. This burning mechanism is designed to create scarcity, which can lead to price appreciation if demand remains constant or increases. The regularity and transparency of these burns have contributed to maintaining a high and stable price for BNB.

Another example can be found in the context of USD Tether (USDT), where token burning plays a crucial role in maintaining the price peg to the USD. Tether achieves price stability by issuing or redeeming tokens based on the corresponding deposit or withdrawal of USD from a bank account. When demand for Tether increases and the price deviates from the one-to-one peg, arbitrageurs buy Tether at a discount and sell it back to Tether Limited in exchange for USD, which then results in the redemption and "burning" of Tether tokens to reduce the supply. This mechanism allows Tether to remove excess tokens from circulation, helping to restore the peg. Conversely, when the price rises above the peg, new tokens are issued. Thus, the controlled burning and issuance of tokens are essential for maintaining the peg and ensuring the price stability of Tether (Ciriello, 2021).

Token Minting

Token minting is another tokenomics mechanism that involves creating new tokens and introducing them into circulation. This process can serve various purposes, such as incentivizing network participation, rewarding contributors, or maintaining the stability of a token's value. Minting can be programmed to occur at regular intervals, triggered by specific events, or

controlled by governance decisions made by the community. The rate and conditions of token minting significantly impact a token's supply dynamics, which in turn affects its value and utility within the ecosystem.

For example, in the MakerDAO system, users can mint new DAI tokens by depositing cryptocurrency collateral into a Maker Vault. The minting process is directly tied to the value of the collateral, ensuring that each DAI token is backed by assets of greater value. For example, if a user deposits 150 USD worth of Ethereum, they may be able to mint up to 100 DAI, maintaining a collateralization ratio of 150%. This mechanism allows for a dynamic and responsive token supply that adjusts based on user demand and market conditions, while simultaneously maintaining the stability of DAI's value relative to the USD.

The distribution of tokens is critical in establishing alignment among stakeholders, especially during the early-stage evolution of a blockchain-based community. It allows for the creation of a more decentralized governance structure, promotes liquidity, and incentivizes early adopters. However, it also presents challenges, particularly around ensuring equitable distribution and avoiding concentration of tokens among a small group of participants. Protocols often employ strategies such as community rewards, liquidity mining programs, or vesting schedules with controlled token unlocks to address these issues. Token unlocks are scheduled events in which previously locked tokens (according to a vesting schedule) are gradually released into circulation. These unlocks can be tied to milestones, investor agreements, or incentives, and are designed to manage the token supply while supporting liquidity. However, poorly timed or overly aggressive token unlocks can disrupt market dynamics, leading to price volatility as large quantities of tokens are suddenly released. In the Steem protocol, for example, tokens such as Steem Power (SP) are subject to a vesting schedule. Users commit their STEEM tokens to a 13-week vesting period, during which these tokens cannot be transferred or sold. This ensures that participants maintain a long-term commitment to the platform, promoting stability and encouraging them to contribute positively to the ecosystem during this period. In Steem's case, SP holders also gain voting power and influence proportional to their vested holdings, which incentivizes thoughtful governance.

Governance through Token-Based Voting

A crucial aspect in tokenomics design is how governance tokens are designed to function within a protocol, how they distribute power, align incentives, and facilitate decentralized decision-making. Governance tokens empower token holders to participate in the decision-making processes that shape the protocol. Unlike traditional governance models where centralized entities hold control, governance tokens distribute this authority among token holders, creating a system that reflects the collective will of its community under certain conditions explored in the literature (e.g., Ellinger et al., 2024). This relatively more decentralized governance ensures that more stakeholders have a say in the protocol's future, aligning economic incentives with the long-term success of the system. More decentralized governance typically operates through a token-based voting system, where governance tokens grant holders the right to propose and vote on changes to the protocol. The weight of a token holder's vote is typically, but not always,

proportional to the number of tokens they hold, incentivizing active participation by those with a vested interest in the protocol's success and creating "skin in the game." Token holders are tasked with making decisions on a range of issues, including protocol upgrades, risk management adjustments, and changes to fee structures.

MakerDAO, for instance, employs MKR tokens to give its community control over critical aspects of its system. MKR holders vote on decisions such as the collateral types that back the DAI stablecoin, rates, and risk parameters. This governance structure is integral to maintaining DAI's stability and peg to the USD. For example, the DAI Savings Rate (DSR) allows any DAI holder to earn savings automatically and natively by locking their DAI into the DSR smart contract in the Maker Protocol. If the market price of DAI is above 1 USD, MKR holders can choose to gradually decrease the DAI savings rate (DSR), which will reduce demand and should reduce the market price of DAI toward the 1 USD target price. In turn, if the market price of DAI is below 1 USD, MKR holders can choose to gradually increase the DSR, which will stimulate demand and should increase the market price of DAI toward the 1 USD target price. By involving MKR holders in these decisions, MakerDAO ensures that those who stand to benefit or lose from the protocol's stability are directly responsible for its governance.

Aave operates a similar decentralized governance model through its AAVE token, which allows holders to propose and vote on protocol changes. This includes decisions such as introducing new assets, modifying liquidity parameters, or adjusting risk levels. Additionally, Aave integrates its governance model with its Safety Module, where AAVE holders can stake their tokens as insurance against potential protocol failures. This feature encourages token holders to make decisions that prioritize the security and sustainability of the protocol, as their staked assets are directly tied to the protocol's health.

Token-based governance has several benefits. It ensures that the stakeholders, who have a vested interest in the success of the platform, influence decisions. It also decentralizes power, ensuring that no single entity can control the platform's direction. In Aave, governance also extends to voting on the allocation of liquidity, the implementation of new features, and interest rate strategies.

However, governance through tokens also introduces challenges. Concentration of tokens among a few large holders (commonly known as "whales") can lead to centralized control, undermining the core principle of decentralization. As a result, many protocols have implemented systems that balance the influence of large holders while maintaining a participatory process for smaller stakeholders.

Incentive Structures: Staking and Yield Generation Mechanisms

Staking plays a crucial role in the economic model of tokens, incentivizing token holders to lock their assets within the protocol in exchange for rewards. This mechanism not only supports the security of the network but also generates yields for participants.

Lido, for example, operates a liquid staking model for Ethereum 2.0, allowing users to stake ETH and receive stETH tokens in return. These stETH tokens represent the user's staked ETH and accumulate staking rewards. One of the major advantages of liquid staking is that users can participate in staking without locking up their capital, as the stETH tokens remain tradable and usable within DeFi applications. This increases liquidity in the ecosystem and provides more flexibility for stakers.

Compound and Aave take a different approach by offering yield generation through lending and borrowing. In Compound, users deposit assets into liquidity pools and receive interest, which is algorithmically determined based on the supply and demand for each asset. Borrowers take loans from these pools by using other assets as collateral, and the interest they pay funds the yields for the liquidity providers. Similarly, Aave uses an interest rate model where users can borrow against their assets, and interest rates fluctuate based on market conditions.

Staking and yield generation are powerful mechanisms that fuel the growth of decentralized finance (DeFi) ecosystems. These mechanisms provide a dual benefit: they incentivize user participation by offering rewards, and they ensure the availability of liquidity for borrowing, lending, or trading. However, they are not without risks, especially in terms of liquidity volatility and asset devaluation. Protocols manage these risks through mechanisms such as liquidation thresholds, over-collateralization, and dynamic interest rates.

Risk Management and Collateralization Mechanisms

A critical aspect of tokenomics is the implementation of risk management and collateralization mechanisms. These mechanisms ensure the stability and security of the protocol, protecting it from external shocks and minimizing the risk of loss for participants. Many blockchain-based collectives rely heavily on collateral to secure the network and mitigate the risk of defaults, particularly in lending and borrowing protocols.

One of the primary risk management strategies used is over-collateralization. In protocols like MakerDAO, users must deposit collateral assets that exceed the value of the DAI stablecoin they wish to mint. This over-collateralization ensures that there is sufficient backing for the DAI issued, protecting the system from insolvency in the event of a market downturn. If the value of the collateral falls below a certain threshold, the system automatically liquidates the collateral to repay the DAI debt, protecting the protocol from systemic risk. In contrast, Tether, a popular stablecoin, has faced accusations of under-collateralization, with critics questioning whether it holds sufficient reserves to back the USDT tokens it issues (Eichengreen, 2019). This discrepancy highlights the importance of maintaining robust collateralization mechanisms to ensure stability and trust in decentralized financial systems.

Similarly, in Compound and Aave, users are required to provide collateral before borrowing assets from the protocol. The amount of collateral required is determined by the asset's collateral factor, which reflects the asset's risk profile. More volatile assets have lower collateral factors, requiring users to post more collateral to secure a loan. This system ensures that the protocol remains solvent even if the value of the collateralized assets fluctuates.

Aave also introduces innovative risk management tools such as “flash loans,” which allow users to borrow assets without collateral as long as the loan is repaid within the same transaction. This unique feature enables users to engage in arbitrage opportunities and liquidity rebalancing without risking the protocol’s stability. Flash loans provide a novel way to manage liquidity risk and ensure that the protocol remains robust, even in times of market volatility.

Moreover, MakerDAO’s stability fee mechanism plays a significant role in maintaining the peg of the DAI stablecoin to the USD. The stability fee is a variable interest rate charged on DAI loans, incentivizing users to repay their loans during periods of high demand for DAI, thereby reducing the supply of DAI in circulation and stabilizing its price.

Through these risk management and collateralization mechanisms, DAOs can maintain the security and stability of their ecosystems, ensuring that participants’ assets are protected while minimizing the risk of default or insolvency. These mechanisms are crucial in lending and borrowing protocols, where the risk of default can have cascading effects on the broader network.

Conclusion

In conclusion, tokenomics plays a crucial role in shaping the economic landscape of blockchain networks and decentralized applications. By carefully designing token distribution, governance mechanisms, incentive structures, and risk management protocols, blockchain-based collectives, communities, and projects try to create sustainable ecosystems that align participant interests with long-term network health.

References and selected further readings

- Ciriello, R. F. (2021). Tokenized index funds: A blockchain-based concept and a multidisciplinary research framework. *International Journal of Information Management*, 61, 102400.
- Dixon, C. (2024). *Read Write Own: Building the Next Era of the Internet*, Random House.
- Eichengreen, B. (2019). From Commodity to Fiat and now Crypto: What does History tell us? NBER Working Paper 25426. Retrieved 3 October 2024 from <http://www.nber.org/papers/w25426>
- Ellinger, E. W., Gregory, R. W., Mini, T., Widjaja, T., & Henfridsson, O. (2024). Skin in the game: The transformational potential of decentralized autonomous organizations. *MIS Quarterly*, 48(1), 245-272.
- Gregory, R. W., Beck, R., Henfridsson, O., Yaraghi, N. (2024). Cooperation Among Strangers: Algorithmic Enforcement of Reciprocal Exchange with Blockchain-Based Smart Contracts. *Academy of Management Review*. forthcoming.
- Lesche, P., Sandner, P., & Treiblmaier, H. (2022). Implications of the token economy: A taxonomy and research agenda. *Blockchains and the token economy: Theory and practice*, 1-30.
- Sockin, M., & Xiong, W. (2023). Decentralization through Tokenization. *The Journal of Finance*, 78(1), 247-299.